Database Tables

Asset Management

Simple tracking of company assets and who is using them.

> **Assets** - List of all company assets (computers, furniture, equipment, etc.)
> **Asset_Types** - Categories of assets (e.g., "Laptop," "Chair," "Monitor")
> **Accessories** - List of accessory items that can be tracked
> **Accessory_Types** - Categories of accessories (e.g., "Mouse," "Keyboard," "Headset")
> **Asset_Accessory_Links** - Link accessories to assets (e.g., mouse bundled with desktop)
> **Accessory_Assignments** - Track which accessories are assigned to employees (standalone only)
> **Asset_Images** - Photos of assets
> **Asset_Comments** - Notes and comments about assets with optional photos

People & Places

Track who has what and where things are located.

> **Employees** - List of all employees
> **Departments** - Company departments
> **Branches** - Company branch locations
> **Asset_Assignments** - Track who has which asset (current and history)

Simple Tracking

Basic records of what happens to assets.

> **Asset_Reports** - Simple log when something happens to an asset
> **Asset_Disposal** - Record when asset is retired/lost/disposed
> **Audit_Log** - Track all changes made in the system

Supply Management

Track office supplies and who requests them.

> **Supplies** - List of all office supplies
> **Categories** - Groups supplies (e.g., "Pens," "Paper," "Printer Supplies")
> **Inventory_Stock** - How many supplies are in stock
> **Stock_Locations** - Where supplies are stored
> **Supply_Requests** - Track who requests supplies and how often
> **Stock_Transactions** - Record when supplies come IN or go OUT

System Administration

User accounts and access control.

> **Users** - System users and their login credentials
> **Roles** - User roles and permission levels
> **Contracts** - Asset contracts, warranties, and agreements

Processes

1. **Add New Asset**
    1. Create record in **Assets** table with basic info
    2. Add manufacturer and model if known (optional for old assets)
    3. Upload photo if available
    4. Set status as "Available"
2. **Assign Asset to Employee**
    1. Select asset and employee
    2. Select branch where asset will be used
    3. Create record in **Asset_Assignments**
    4. Update asset status to "Assigned"
3. **Assign Accessories**
    1. **Option A: Assign Independent Accessory**
        - Select employee and specific accessory
        - Create record in **Accessory_Assignments**
        - Accessory is tracked separately
    2. **Option B: Accessory Bundled with Asset**
        - Accessory is linked to an asset via **Asset_Accessory_Links**
        - When the asset is assigned, all bundled accessories go with it
        - No separate accessory assignment needed
4. **Transfer Asset to Different Employee**
    1. Close current assignment (add return date)
    2. Create new assignment for new employee
    3. System keeps history of all assignments
5. **Report Issue with Asset**
    1. Create simple report in **Asset_Reports**
    2. Describe what happened

    3. Note what was done about it (optional)

6. **Retire/Dispose Asset**
    1. Create record in **Asset_Disposal**
    2. Write reason for disposal
    3. Set asset status to "Disposed" or "Lost"
    4. Close any open assignment
7. **Request Supplies**
    1. Employee requests supplies
    2. Create record in **Supply_Requests**
    3. System tracks who requests, what they request, and how often
8. **Give Out Supplies**
    1. Record transaction in **Stock_Transactions** as "OUT"
    2. Reduce stock count in **Inventory_Stock**
    3. Note reason: Dispensed, Lost, or Damaged
9. **Restock Supplies**
    1. Record transaction in **Stock_Transactions** as "IN"
    2. Increase stock count in **Inventory_Stock**

Data Dictionary

I. Asset Management

Assets

Field Name	Data Type	Notes	Description
Asset_ID	INT	Primary Key, Auto-increment	Unique ID for each asset
Asset_Tag	VARCHAR(50)	Unique, Required	Company's asset tag/sticker number
Asset_Name	VARCHAR(150)	Required	Name of the asset (e.g., "Dell Laptop")
Description	TEXT	Required	Detailed description of the asset

Field Name	Data Type	Notes	Description
Manufacturer	VARCHAR(100)	Optional	Manufacturer name (e.g., "Dell", "HP")
Model	VARCHAR(100)	Optional	Model number/name (e.g., "Latitude 5420")
Serial_Number	VARCHAR(100)	Optional	Manufacturer's serial number (if available)
Asset_Type_ID	INT	FK -> Asset_Types	Type/category of asset
Purchase_Date	DATE	Required	When asset was acquired
Cost	DECIMAL(10,2)	Optional	How much it cost (if known)
Status	VARCHAR(50)	Required	Current status: Available, Assigned, In Repair, Disposed, Lost
Current_Branch_ID	INT	FK -> Branches	Which branch currently has this asset

Note: Manufacturer and Model are optional fields to accommodate older assets where this information may not be available or recorded.

Asset_Types

Field Name	Data Type	Notes	Description
Type_ID	INT	Primary Key	Unique ID
Type_Name	VARCHAR(100)	Unique, Required	Category name (e.g., "Laptop", "Chair")

Accessories

Field Name	Data Type	Notes	Description
Accessory_ID	INT	Primary Key	Unique ID
Accessory_Name	VARCHAR(150)	Required	Name (e.g., "Logitech Mouse M185")
Accessory_Type_ID	INT	FK -> Accessory_Types, Optional	Type/category of accessory
Description	TEXT		Details about the accessory
Serial_Number	VARCHAR(100)	Optional	Serial number if applicable

Field Name	Data Type	Notes	Description
Status	VARCHAR(50)	Required	Available, Assigned, Bundled, Lost, Damaged
Current_Branch_ID	INT	FK -> Branches	Which branch currently has this accessory

Note: This serves as the master list of ALL accessories in the company, whether they're bundled with assets or standalone.

Accessory_Types

Field Name	Data Type	Notes	Description
Type_ID	INT	Primary Key	Unique ID
Type_Name	VARCHAR(100)	Unique, Required	Category name (e.g., "Mouse", "Keyboard", "Headset", "Webcam")

Asset_Accessory_Links

Field Name	Data Type	Notes	Description
Link_ID	INT	Primary Key	Unique ID
Asset_ID	INT	FK -> Assets	Which asset this accessory is bundled with
Accessory_ID	INT	FK -> Accessories	Which accessory
Link_Date	DATE	Required	When bundled together
Unlink_Date	DATE	Optional	When unbundled (empty if still bundled)

Note: When an accessory is bundled with an asset, its status becomes "Bundled". When the asset is assigned to someone, the bundled accessories automatically go with it. Accessories can be unbundled later if needed.

Accessory_Assignments

Field Name	Data Type	Notes	Description
Assignment_ID	INT	Primary Key	Unique ID
Employee_ID	INT	FK -> Employees	Who has it

Field Name	Data Type	Notes	Description
Accessory_ID	INT	FK -> Accessories	Which standalone accessory
Branch_ID	INT	FK -> Branches, Required	Which branch
Assignment_Date	DATE	Required	When assigned
Return_Date	DATE	Optional	When returned (empty if still assigned)
Notes	TEXT		Any notes about the assignment

Note: This table is only for standalone accessories that are NOT bundled with assets. Bundled accessories follow the asset assignment automatically.

Asset_Images

Field Name	Data Type	Notes	Description
Image_ID	INT	Primary Key	Unique ID
Asset_ID	INT	FK -> Assets	Which asset this photo belongs to
Image_Path	VARCHAR(255)	Required	Where the photo is stored
Upload_Date	DATETIME	Required	When photo was added
Uploaded_By	INT	FK -> Employees	Who uploaded the photo

Asset_Comments

Field Name	Data Type	Notes	Description
Comment_ID	INT	Primary Key	Unique ID
Asset_ID	INT	FK -> Assets	Which asset
Comment_Text	TEXT	Required	The comment/note
Comment_Date	DATETIME	Required	When comment was made
Created_By	INT	FK -> Employees	Who made the comment
Photo_Path	VARCHAR(255)	Optional	Attached photo (if any)

II. People & Places

Employees

Field Name	Data Type	Notes	Description
Employee_ID	INT	Primary Key	Unique ID
First_Name	VARCHAR(50)	Required	First name
Last_Name	VARCHAR(50)	Required	Last name
Email	VARCHAR(100)	Unique, Required	Email address
Department_ID	INT	FK -> Departments	Which department
Branch_ID	INT	FK -> Branches	Which branch
Status	VARCHAR(20)	Required	Active or Inactive

Departments

Field Name	Data Type	Notes	Description
Dept_ID	INT	Primary Key	Unique ID
Dept_Name	VARCHAR(100)	Unique, Required	Department name

Branches

Field Name	Data Type	Notes	Description
Branch_ID	INT	Primary Key	Unique ID
Branch_Name	VARCHAR(100)	Unique, Required	Branch name
Address	TEXT		Branch address

Asset_Assignments

Field Name	Data Type	Notes	Description
Assignment_ID	INT	Primary Key	Unique ID
Asset_ID	INT	FK -> Assets	Which asset
Employee_ID	INT	FK -> Employees	Who has it
Branch_ID	INT	FK -> Branches, Required	Which branch
Assignment_Date	DATE	Required	When assigned
Return_Date	DATE	Optional	When returned (empty if still assigned)
Notes	TEXT		Any notes about the assignment

Note: System automatically tracks assignment history - you can see where an asset was assigned before by looking at all records for that Asset_ID ordered by date.

III. Simple Asset Tracking

Asset_Reports

Field Name	Data Type	Notes	Description
Report_ID	INT	Primary Key	Unique ID
Asset_ID	INT	FK -> Assets	Which asset
Report_Date	DATETIME	Required	When reported
Reported_By	INT	FK -> Employees	Who reported it
What_Happened	TEXT	Required	Description of what happened
What_Was_Done	TEXT	Optional	What action was taken (if any)

Note: Simple tracking of what happens to assets - no complex maintenance schedules, just basic reporting.

Asset_Disposal

Field Name	Data Type	Notes	Description
Disposal_ID	INT	Primary Key	Unique ID
Asset_ID	INT	FK -> Assets	Which asset
Disposal_Date	DATE	Required	When disposed
Reason	TEXT	Required	Description of why it was disposed

Note: Simple disposal tracking - just record the reason/description without complex categorization.

Audit_Log

Field Name	Data Type	Notes	Description
Log_ID	INT	Primary Key	Unique ID
Table_Name	VARCHAR(100)	Required	Which table was affected
Record_ID	INT	Required	Which record was changed
Action_Type	VARCHAR(50)	Required	Created, Updated, Deleted
Old_Value	TEXT	Optional	Previous value (for updates)
New_Value	TEXT	Optional	New value (for updates)
Changed_By	INT	FK -> Users	Who made the change
Changed_At	DATETIME	Required	When the change was made

Note: Complete audit trail of all system changes for accountability and tracking.

IV. Supply Management

Supplies

Field Name	Data Type	Notes	Description
Supply_ID	INT	Primary Key	Unique ID

Field Name	Data Type	Notes	Description
Supply_Name	VARCHAR(150)	Unique, Required	Name of supply item
Category_ID	INT	FK -> Categories	What category
Unit_of_Measure	VARCHAR(20)	Required	How it's counted (Box, Each, Pack, etc.)
Is_Frequently_Lost	BOOLEAN	Default: False	Flag items that go missing often (like scissors)

Note: Track frequently lost items with the Is_Frequently_Lost flag for better monitoring and reporting.

Categories

Field Name	Data Type	Notes	Description
Cat_ID	INT	Primary Key	Unique ID
Cat_Name	VARCHAR(100)	Unique, Required	Category name

Inventory_Stock

Field Name	Data Type	Notes	Description
Stock_ID	INT	Primary Key	Unique ID
Supply_ID	INT	FK -> Supplies	Which supply
Stock_Location_ID	INT	FK -> Stock_Locations	Where stored
Current_Count	INT	Required, Default: 0	How many in stock now
Reorder_Level	INT	Optional	Low stock alert level (if set)
Unit_Cost	DECIMAL(10,2)	Optional	Cost per unit (if known)

Stock_Locations

Field Name	Data Type	Notes	Description
Location_ID	INT	Primary Key	Unique ID
Location_Name	VARCHAR(100)	Unique, Required	Name of storage area

Field Name	Data Type	Notes	Description
Branch_ID	INT	FK -> Branches	Which branch

Supply_Requests

Field Name	Data Type	Notes	Description
Request_ID	INT	Primary Key	Unique ID
Employee_ID	INT	FK -> Employees	Who is requesting
Department_ID	INT	FK -> Departments	Which department
Request_Date	DATETIME	Required	When requested
Status	VARCHAR(20)	Required	Pending, Approved, Fulfilled, Rejected

Supply_Request_Items

Field Name	Data Type	Notes	Description
Item_ID	INT	Primary Key	Unique ID
Request_ID	INT	FK -> Supply_Requests	Which request
Supply_ID	INT	FK -> Supplies	What supply
Quantity_Requested	INT	Required	How many requested
Notes	TEXT		Any special notes

Stock_Transactions

Field Name	Data Type	Notes	Description
Transaction_ID	INT	Primary Key	Unique ID
Supply_ID	INT	FK -> Supplies	Which supply
Stock_Location_ID	INT	FK -> Stock_Locations	Which storage location
Transaction_Type	VARCHAR(10)	Required	"IN" or "OUT"

Field Name	Data Type	Notes	Description
Reason	VARCHAR(50)	Required for OUT	If OUT: Dispensed, Lost, Damaged, or other reason
Quantity	INT	Required	How many
Transaction_Date	DATETIME	Required	When
Employee_ID	INT	FK -> Employees	Who (for OUT transactions)
Department_ID	INT	FK -> Departments	Which department (for OUT)
Request_ID	INT	FK -> Supply_Requests	Link to request (if from request)
Receipt_Path	VARCHAR(255)	Optional	Receipt/proof (for IN)
Notes	TEXT		Any notes

V. System Administration

Users

Field Name	Data Type	Notes	Description
User_ID	INT	Primary Key	Unique ID
Username	VARCHAR(50)	Unique, Required	Login username
Password_Hash	VARCHAR(255)	Required	Encrypted password
Is_Active	BOOLEAN	Required, Default: True	Account active status
Last_Login	DATETIME	Optional	Last login timestamp
Created_At	DATETIME	Required	When account was created
Updated_At	DATETIME	Required	Last account update
Role_ID	INT	FK -> Roles	User role/permission level
Employee_ID	INT	FK -> Employees	Link to employee record

Roles

Field Name	Data Type	Notes	Description
Role_ID	INT	Primary Key	Unique ID
Role_Name	VARCHAR(100)	Unique, Required	Role name (Admin, Manager, User, etc.)
Description	TEXT		What this role can do
Is_Active	BOOLEAN	Required, Default: True	Role active status
Created_At	DATETIME	Required	When role was created

Contracts

Field Name	Data Type	Notes	Description
Contract_ID	INT	Primary Key	Unique ID
Contract_Type	VARCHAR(100)	Required	Type (Warranty, Lease, Service, etc.)
Start_Date	DATE	Required	Contract start date
End_Date	DATE	Required	Contract end date
Coverage_Details	TEXT		What's covered
Contract_File_Path	VARCHAR(255)	Optional	Path to contract document
Asset_ID	INT	FK -> Assets	Which asset this contract covers
Vendor_Name	VARCHAR(150)	Optional	Vendor/shop name

Note: Vendor information is kept minimal - just the shop/vendor name for easy tracking.

Key Features

- **Simple Asset Tracking** - Know what you have, who has it, and where it is
- **Manufacturer & Model Info** - Track manufacturer and model details (optional for older assets)
- **Flexible Accessory Management** - Accessories can work standalone OR be bundled with assets
- **Asset Bundling** - Link accessories to assets (e.g., mouse & keyboard bundled with desktop)

- **Accessory Master List** - Complete inventory of all accessories, whether bundled or standalone
- **Assignment History** - See complete history of who had an asset or accessory
- **Photos** - Attach photos to assets and comments
- **Branch Operations** - Track which branch has what
- **Simple Reporting** - Basic tracking of what happens to assets without complex maintenance schedules
- **Audit Trail** - Complete log of all system changes for accountability
- **Supply Tracking** - Track supplies with IN/OUT transactions
- **Frequently Lost Items** - Flag and monitor items that go missing often
- **Receipt Storage** - Keep receipts for supplies purchased
- **No Complex Procurement** - Just basic tracking, no purchase orders or approvals

What This System Does NOT Do

- Purchase requisitions and approvals
- Vendor management and RFQs
- Complex budgeting and cost centers
- Centralized procurement workflows
- Multi-level approval chains
- Depreciation calculations
- Detailed maintenance schedules

This is a **simple tracking system** focused on:

> Who has what asset
> Where assets are located
> Who requests supplies and how often
> Basic stock levels
> Flexible accessory tracking (standalone or bundled with assets)